

4(a) The increase in internal energy of a system is equal to the sum of the heat supplied to the system and the work done on the system.

(b)(i) AB: volume is compressed, so $(W_{on})_{AB}$ is positive.
Adiabatic: $Q_{AB} = 0$

$$\begin{aligned}\Delta U_{AB} &= Q_{AB} + (W_{on})_{AB} \\ &= 0 + 10 \\ &= 10 \text{ J [A1]} \quad (\text{Increase})\end{aligned}$$

(ii) For a complete cycle, $\Delta U_{AB} + \Delta U_{BC} + \Delta U_{CD} + \Delta U_{DA} = 0$

$$\begin{aligned}\Delta U_{AB} + [Q_{BC} + (W_{on})_{BC}] + [Q_{CD} + (W_{on})_{CD}] + [Q_{DA} + (W_{on})_{DA}] &= 0 \\ 10 + [70 + 0] + [0 - 25] + [Q_{DA} + 0] &= 0 \\ Q_{DA} &= -55 \text{ J} \\ \text{So 55 J of heat leaves the gas.}\end{aligned}$$

(iii) Efficiency = $\frac{\text{Net useful work done by gas}}{\text{Heat input}}$

$$\begin{aligned}&= \frac{[(W_{by})_{CD} - (W_{by})_{AB}]}{Q_{BC}} \\ &= \frac{25 - 10}{70} \\ &= 21 \%\end{aligned}$$

(iv) Useful power output
= (Net useful work done by gas per cycle) x (No. of cycles per second)

$$\begin{aligned}&= (25 - 10) \times \frac{3000}{60} \\ &= 750 \text{ W}\end{aligned}$$

5(a) Lenz's law states that the induced emf and hence current in conductor will be such